

Git & GitHub Command Handbook

Quick reference guide for developers and students.

Git Setup

git --version — Check Git version

git config --global user.name 'Your Name' — Set username

git config --global user.email 'you@example.com' — Set email

git config --list — View configuration

Repository Creation

git init — Initialize repository

git clone — Clone repository

Daily Workflow

git status — Check status

git add . — Stage all files

git add — Stage specific file

git commit -m 'message' — Create commit

git push origin main — Push changes

git pull origin main — Pull changes

Branching

git branch — List branches

git branch feature — Create branch

git checkout feature — Switch branch

git checkout -b feature — Create and switch

git merge feature — Merge branch

git branch -d feature — Delete branch

History & Logs

git log — View commit history

git log --oneline — Compact history

git diff — Show changes

git show — View commit details

Undo & Recovery

git restore — Discard changes

git reset --soft HEAD~1 — Undo last commit

git reset --hard HEAD~1 — Remove commit and changes

git revert — Reverse a commit safely

GitHub Commands

gh auth login — Authenticate GitHub CLI

gh repo create — Create repository

gh repo clone owner/repo — Clone repository

gh pr create — Create pull request

gh pr list — List pull requests

gh issue list — List issues

Useful Aliases

git config --global alias.st status — git st

git config --global alias.co checkout — git co

git config --global alias.br branch — git br

git config --global alias.lg 'log --oneline --graph --all' — git lg

Common Git Workflow

git clone

git checkout -b feature-branch

git add .

git commit -m 'Implemented feature'

git push origin feature-branch

Create Pull Request on GitHub